Data Cleaning

2026-10-30

Set your Working Directory from Session to the Repository Folder.

Upload the two databases.

```
q_data <- read_xpt("data/ARQ_F.xpt")
x_data <- read_xpt("data/ARX_F.xpt")
```

```
dim(q_data)
```

```
## [1] 5106  125
```

```
dim(x_data)
```

```
## [1] 5001   13
```

```
head(q_data)
```

```
## # A tibble: 6 x 125
##    SEQN ARQ010 ARQ020A ARQ020B ARQ020C ARQ020D ARQ020E ARQ020F ARQ020G ARQ021AA
##   <dbl>  <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>    <dbl>
## 1 51624      1      NA      NA      NA       4       5      NA      NA       NA
## 2 51628      1      NA      NA       3       4       5       6      NA       NA
## 3 51629      2      NA      NA      NA      NA      NA      NA      NA       NA
## 4 51630      1      NA      NA      NA      NA       5      NA      NA       NA
## 5 51637      2      NA      NA      NA      NA      NA      NA      NA       NA
## 6 51643      2      NA      NA      NA      NA      NA      NA      NA       NA
## # i 115 more variables: ARQ021BA <dbl>, ARQ021CA <dbl>, ARQ022A <dbl>,
## #   ARQ022AA <dbl>, ARQ023GA <dbl>, ARQ023QA <dbl>, ARQ024A <dbl>,
## #   ARQ025A <dbl>, ARD026A <dbl>, ARQ021AB <dbl>, ARQ021BB <dbl>,
## #   ARQ021CB <dbl>, ARQ022B <dbl>, ARQ022AB <dbl>, ARQ023GB <dbl>,
## #   ARQ023QB <dbl>, ARQ024B <dbl>, ARQ025B <dbl>, ARD026B <dbl>,
## #   ARQ021AC <dbl>, ARQ021BC <dbl>, ARQ021CC <dbl>, ARQ022C <dbl>,
## #   ARQ022AC <dbl>, ARQ023GC <dbl>, ARQ023QC <dbl>, ARQ024C <dbl>, ...
```

```
head(x_data)
```

```
## # A tibble: 6 x 13
##    SEQN ARDEXSTS ARXO2WD ARIO2WD ARXCCIN ARICCIN ARXCCEX ARICCEX ARDDINEX
##   <dbl>    <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>    <dbl>
## 1 51624        2       0      NA     103      NA    98.6      NA     4.40
## 2 51628        1       0      NA     110.     NA   107        NA     2.5
```

```
## 3 51629        1     0        NA    103.      NA    98        NA     5.2
## 4 51630        1     0        NA    104.      NA   100.       NA     4.20
## 5 51643        1     0        NA    111.      NA   104.       NA     6.8
## 6 51645        2     3.2      NA     NA        1    NA         1    NA
## # i 4 more variables: ARXXDIST <dbl>, ARIXDIST <dbl>, ARDLFTL <dbl>,
## #   ARILFTL <dbl>
```

A few checks need to be ran to determine if the two databases can be combined for the study.

Q_Data: A dataset featuring questionnaire responses for several experiences and attributes pertaining to arthritis.
X_Data: Arthritis physical measurements of the same patients that also covers various health data.

The idea is to combine the two to evaluate patients with full coverage for said data, since both inquiries and examinations can be scrutinized for trends and relationships.

```
"SEQN" %in% names(q_data)
```

```
## [1] TRUE
```

```
"SEQN" %in% names(x_data)
```

```
## [1] TRUE
```

```
sum(is.na(q_data$SEQN))
```

```
## [1] 0
```

```
sum(is.na(x_data$SEQN))
```

```
## [1] 0
```

```
sum(duplicated(q_data$SEQN))
```

```
## [1] 0
```

```
sum(duplicated(x_data$SEQN))
```

```
## [1] 0
```

The datasets are essentially sibling datasets. They share the same identifier column 'SEQN', and as checked, the identifiers are complete and unique within each dataset. We also check for potential edge cases, such as potentially redundant participants that make up the observations. Since all returns TRUE or 0, we know these shouldn't be problematic.

```
length(intersect(q_data$SEQN, x_data$SEQN))
```

```
## [1] 5001
```

```r
sum(!q_data$SEQN %in% x_data$SEQN)
```

```
## [1] 105
```

```r
sum(!x_data$SEQN %in% q_data$SEQN)
```

```
## [1] 0
```

Now we check to see if any observations are unique to particular datasets. 5001 participants appear in both datasets, but for the questionnaire, 105 are missing physical observations in x_data. Joining the datasets will exclude 105 observations from the final set, but it should still be a substantial amount of data for analysis.

```r
source("01_funct_DataIntegration.R")

raw_data <- merge_datasets(
  q_data, x_data, "SEQN"
)
```

The function pulled will combine the two datasets with inner_join().

```r
dim(raw_data)
```

```
## [1] 5001  137
```

```r
head(raw_data)
```

```
## # A tibble: 6 x 137
##    SEQN ARQ010 ARQ020A ARQ020B ARQ020C ARQ020D ARQ020E ARQ020F ARQ020G ARQ021AA
##   <dbl>  <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>    <dbl>
## 1 51624      1      NA      NA      NA       4       5      NA      NA       NA
## 2 51628      1      NA      NA       3       4       5       6      NA       NA
## 3 51629      2      NA      NA      NA      NA      NA      NA      NA       NA
## 4 51630      1      NA      NA      NA      NA       5      NA      NA       NA
## 5 51643      2      NA      NA      NA      NA      NA      NA      NA       NA
## 6 51645      2      NA      NA      NA      NA      NA      NA      NA       NA
## # i 127 more variables: ARQ021BA <dbl>, ARQ021CA <dbl>, ARQ022A <dbl>,
## #   ARQ022AA <dbl>, ARQ023GA <dbl>, ARQ023QA <dbl>, ARQ024A <dbl>,
## #   ARQ025A <dbl>, ARD026A <dbl>, ARQ021AB <dbl>, ARQ021BB <dbl>,
## #   ARQ021CB <dbl>, ARQ022B <dbl>, ARQ022AB <dbl>, ARQ023GB <dbl>,
## #   ARQ023QB <dbl>, ARQ024B <dbl>, ARQ025B <dbl>, ARD026B <dbl>,
## #   ARQ021AC <dbl>, ARQ021BC <dbl>, ARQ021CC <dbl>, ARQ022C <dbl>,
## #   ARQ022AC <dbl>, ARQ023GC <dbl>, ARQ023QC <dbl>, ARQ024C <dbl>, ...
```

```r
sum(duplicated(raw_data$SEQN))
```

```
## [1] 0
```

```r
n_distinct(raw_data$SEQN)
```

```
## [1] 5001
```

From a visual examination, all of the columns are present alongside their original patients. The dimension read also supports the 5001 compatible cases. For now it's structurally consistent and the integration was successful. Certainly unclean, but further checks will establish if that renders the database infeasible for research. There's a decent chance there's too many holes to make meaningful observations.

Adding Demographics and Body Measures

Spinal mobility is related to age, sex and height, so a fair comparison between participants needs those variables. Neither arthritis dataset contains them, so we add two more NHANES 2009-2010 datasets. `DEMO_F.xpt` holds age, sex, race/ethnicity, pregnancy status and the survey design variables, and `BMX_F.xpt` holds height, weight and BMI.

```r
demo_data <- read_xpt("data/DEMO_F.xpt")
body_data <- read_xpt("data/BMX_F.xpt")

dim(demo_data)
```

```
## [1] 10537    43
```

```r
dim(body_data)
```

```
## [1] 10253    23
```

These datasets cover the whole survey, including children, so they are much larger than our arthritis data. Before merging, we confirm that their IDs are unique and that every participant in `raw_data` appears in both.

```r
sum(duplicated(demo_data$SEQN))
```

```
## [1] 0
```

```r
sum(duplicated(body_data$SEQN))
```

```
## [1] 0
```

```r
sum(!raw_data$SEQN %in% demo_data$SEQN)
```

```
## [1] 0
```

```r
sum(!raw_data$SEQN %in% body_data$SEQN)
```

```
## [1] 0
```

All four results are 0, so joining these datasets to `raw_data` will not remove anyone.

```r
raw_data <- raw_data %>%
  merge_datasets(demo_data, "SEQN") %>%
  merge_datasets(body_data, "SEQN")

dim(raw_data)
```

```
## [1] 5001  201
```

```r
n_distinct(raw_data$SEQN)
```

```
## [1] 5001
```

Cleaning Steps

The merged data now has 201 columns, so we reduce it to the variables that matter for studying pain or stiffness and spinal mobility, and fix how those variables are coded. All of the cleaning functions used below are in `02_funct_DataCleaning.R`.

```r
source("02_funct_DataCleaning.R")
```

Step 1: Check who answered the follow-up questions

Blank values in this survey do not all mean the same thing. Most of them are skip patterns: a participant who reported no chronic pain was never asked about their symptoms, so those values are not truly missing. The table below compares chronic pain (`ARQ010`) with whether the participant reached the symptom questions (`ARQ040`).

```r
raw_data %>%
  mutate(reached_symptom_questions = !is.na(ARQ040)) %>%
  count(ARQ010, reached_symptom_questions)
```

```
## # A tibble: 4 x 3
##   ARQ010 reached_symptom_questions     n
##    <dbl> <lgl>                     <int>
## 1      1 FALSE                        85
## 2      1 TRUE                       1398
## 3      2 FALSE                      3516
## 4      9 FALSE                         2
```

Everyone who answered No (2) skipped these questions, as designed. Among those who answered Yes (1), a small group also skipped them. The next table shows why.

```r
raw_data %>%
  filter(ARQ010 == 1, is.na(ARQ040)) %>%
  summarize(
    participants     = n(),
    marked_hip_only  = sum(!is.na(ARQ020F) & is.na(ARQ020G)),
    marked_ribs_only = sum(is.na(ARQ020F) & !is.na(ARQ020G)),
    marked_both      = sum(!is.na(ARQ020F) & !is.na(ARQ020G))
  )
```

```
## # A tibble: 1 x 4
##   participants marked_hip_only marked_ribs_only marked_both
##          <int>           <int>            <int>       <int>
## 1           85              69               13           3
```

These participants only marked the hip or rib cage on the pain diagram, so the survey did not ask them about their symptoms. We record this in `symptoms_asked` so that they are not confused with participants who answered "No" to a symptom.

```
raw_data <- raw_data %>%
  mutate(symptoms_asked = !is.na(ARQ040))
```

Step 2: Check the exam status

`ARDEXSTS` describes how much of the spinal exam each participant finished. We count the measurements available at each exam status.

```
raw_data %>%
  group_by(ARDEXSTS) %>%
  summarize(
    participants      = n(),
    has_occiput_wall  = sum(!is.na(ARXO2WD)),
    has_chest_exp     = sum(!is.na(ARDDINEX)),
    has_lumbar_flex   = sum(!is.na(ARDLFTL))
  )
```

```
## # A tibble: 3 x 5
##   ARDEXSTS participants has_occiput_wall has_chest_exp has_lumbar_flex
##      <dbl>        <int>            <int>         <int>           <int>
## 1        1         4566             4566          4566            4566
## 2        2          385              372            49              99
## 3        3           50                0             0               0
```

Status 1 participants have all three measurements, status 2 participants have some of them, and status 3 participants have none. We keep status 2 participants, because their available measurements are still valid, and each later analysis will use whoever has the measurement it needs.

Step 3: Keep the variables of interest

We keep the main pain questions, the pain diagram sites, the symptom questions, three related diagnoses, the spinal measurements, demographics, body measures and the survey design variables.

We leave out `ARXXDIST` because it records where the chest was measured and not how much the chest moved. We also leave out the comment columns, such as `ARIO2WD`, because they do not contain measurements.

```
selected_data <- raw_data %>%
  select(
    SEQN,
    symptoms_asked,
    ARQ010,
    ARQ020A:ARQ020G,
```

```
    ARQ040, ARQ050, ARQ060, ARQ070, ARQ073, ARQ080, ARQ100, ARQ110,
    ARQ125C, ARQ125D, ARQ125E,
    ARDEXSTS,
    ARXO2WD, ARXCCIN, ARXCCEX, ARDDINEX, ARDLFTL,
    RIDAGEYR, RIAGENDR, RIDRETH1, RIDEXPRG,
    BMXHT, BMXWT, BMXBMI,
    WTMEC2YR, SDMVPSU, SDMVSTRA
  )

dim(selected_data)
```

```
## [1] 5001   37
```

Step 4: Replace the Refused and Don't Know codes

These are stored as 7 and 9 in the survey columns, which could be mistaken for real answers. We set them to `NA`, and only in the columns where those codes apply. The ID, the pain site columns, the exam measurements and the demographic and body measure columns are not included, since 7 and 9 do not mean Refused or Don't know in those columns.

```
survey_cols <- c(
  "ARQ010", "ARQ040", "ARQ050", "ARQ060", "ARQ070", "ARQ073", "ARQ080",
  "ARQ100", "ARQ110", "ARQ125C", "ARQ125D", "ARQ125E"
)

recoded_data <- recode_special_codes(selected_data, survey_cols)

# Number of refused or don't know answers replaced in each column
colSums(is.na(recoded_data[survey_cols])) -
  colSums(is.na(selected_data[survey_cols]))
```

```
##  ARQ010  ARQ040  ARQ050  ARQ060  ARQ070  ARQ073  ARQ080  ARQ100  ARQ110 ARQ125C
##       2      12      13      13      11       2      10       2       2       8
## ARQ125D ARQ125E
##       3       9
```

Step 5: Rename variables and convert categories to factors

The NHANES column names are hard to read and the categories are stored as numbers. We give each variable a descriptive name and turn the categories into labelled factors, using the codes from the NHANES questionnaire documentation.

```
clean_data <- recoded_data %>%
  transmute(
    id = SEQN,
    symptoms_asked,

    # General pain and the pain diagram
    back_pain = label_yes_no(ARQ010),
    site_neck       = mark_pain_site(ARQ020A, back_pain),
    site_upper_back = mark_pain_site(ARQ020B, back_pain),
```

```
    site_mid_back   = mark_pain_site(ARQ020C, back_pain),
    site_low_back   = mark_pain_site(ARQ020D, back_pain),
    site_buttocks   = mark_pain_site(ARQ020E, back_pain),
    site_hip        = mark_pain_site(ARQ020F, back_pain),
    site_rib_cage   = mark_pain_site(ARQ020G, back_pain),

    # Symptom questions
    morning_stiffness = factor(
      ARQ040,
      levels = c(6, 1, 2, 3, 4, 5),
      labels = c("None", "Under 10 min", "10-30 min", "31-60 min",
                 "1-4 hours", "Over 4 hours"),
      ordered = TRUE
    ),
    pain_during_day = factor(
      ARQ050,
      levels = 1:4,
      labels = c("Increases", "Decreases", "Stays the same", "Varies")
    ),
    pain_with_rest = factor(
      ARQ060,
      levels = 1:5,
      labels = c("Increases", "Decreases", "Stays the same", "Varies",
                 "No pain with rest or sleep")
    ),
    pain_wakes_at_night = label_yes_no(ARQ070),
    wakes_after_4_hours = factor(
      ARQ073,
      levels = 1:3,
      labels = c("Yes", "No", "Sleeps under 4 hours")
    ),
    better_with_exercise = factor(
      ARQ080,
      levels = 1:3,
      labels = c("Yes", "No", "Does not do these activities")
    ),
    buttock_pain_switches = label_yes_no(ARQ100),

    # Other conditions
    heel_pain = label_yes_no(ARQ110),
    ulcerative_colitis = label_yes_no(ARQ125C),
    crohns_disease = label_yes_no(ARQ125D),
    ankylosing_spondylitis = label_yes_no(ARQ125E),

    # Spinal exam, all in centimeters
    exam_status = factor(
      ARDEXSTS,
      levels = 1:3,
      labels = c("Complete", "Partial", "Not done")
    ),
    occiput_wall_cm = ARXO2WD,
    chest_inhale_cm = ARXCCIN,
    chest_exhale_cm = ARXCCEX,
```

```
    chest_expansion_cm = ARDDINEX,
    lumbar_flexion_cm = ARDLFTL,

    # Demographics
    age_years = RIDAGEYR,
    sex = factor(RIAGENDR, levels = 1:2, labels = c("Male", "Female")),
    race_ethnicity = factor(
      RIDRETH1,
      levels = 1:5,
      labels = c("Mexican American", "Other Hispanic", "Non-Hispanic White",
                 "Non-Hispanic Black", "Other or Multi-Racial")
    ),
    pregnancy_status = factor(
      RIDEXPRG,
      levels = 1:3,
      labels = c("Pregnant", "Not pregnant", "Could not be determined")
    ),

    # Body measures
    height_cm = BMXHT,
    weight_kg = BMXWT,
    bmi = BMXBMI,

    # Survey design variables, needed for national estimates
    exam_weight = WTMEC2YR,
    survey_psu = SDMVPSU,
    survey_stratum = SDMVSTRA
  )

glimpse(clean_data)
```

```
## Rows: 5,001
## Columns: 37
## $ id                     <dbl> 51624, 51628, 51629, 51630, 51643, 51645, 51647~
## $ symptoms_asked         <lgl> TRUE, TRUE, FALSE, TRUE, FALSE, FALSE, FALSE, F~
## $ back_pain              <fct> Yes, Yes, No, Yes, No, No, No, No, No, No, No, ~
## $ site_neck              <lgl> FALSE, FALSE, NA, FALSE, NA, NA, NA, NA, NA, NA~
## $ site_upper_back        <lgl> FALSE, FALSE, NA, FALSE, NA, NA, NA, NA, NA, NA~
## $ site_mid_back          <lgl> FALSE, TRUE, NA, FALSE, NA, NA, NA, NA, NA, NA,~
## $ site_low_back          <lgl> TRUE, TRUE, NA, FALSE, NA, NA, NA, NA, NA, NA, ~
## $ site_buttocks          <lgl> TRUE, TRUE, NA, TRUE, NA, NA, NA, NA, NA, NA, N~
## $ site_hip               <lgl> FALSE, TRUE, NA, FALSE, NA, NA, NA, NA, NA, NA,~
## $ site_rib_cage          <lgl> FALSE, FALSE, NA, FALSE, NA, NA, NA, NA, NA, NA~
## $ morning_stiffness      <ord> 1-4 hours, Under 10 min, NA, 10-30 min, NA, NA,~
## $ pain_during_day        <fct> Stays the same, Increases, NA, Decreases, NA, N~
## $ pain_with_rest         <fct> Stays the same, Increases, NA, Increases, NA, N~
## $ pain_wakes_at_night    <fct> Yes, Yes, NA, Yes, NA, NA, NA, NA, NA, NA, NA, ~
## $ wakes_after_4_hours    <fct> Yes, Sleeps under 4 hours, NA, Yes, NA, NA, NA,~
## $ better_with_exercise   <fct> No, No, NA, Yes, NA, NA, NA, NA, NA, NA, NA, NA~
## $ buttock_pain_switches  <fct> No, No, NA, Yes, NA, NA, NA, NA, NA, NA, NA, NA~
## $ heel_pain              <fct> No, No, No, No, No, No, No, No, No, No, No, No,~
## $ ulcerative_colitis     <fct> No, No, No, No, No, No, No, No, No, No, No, No,~
## $ crohns_disease         <fct> No, No, No, No, No, No, No, No, No, No, No, No,~
```

```
## $ ankylosing_spondylitis <fct> No, No, No, No, No, No, No, No, No, No, No, No,~
## $ exam_status            <fct> Partial, Complete, Complete, Complete, Complete~
## $ occiput_wall_cm        <dbl> 0.0, 0.0, 0.0, 0.0, 0.0, 3.2, 0.0, 0.0, 0.0, 0.~
## $ chest_inhale_cm        <dbl> 103.0, 109.5, 103.2, 104.5, 111.3, NA, 85.9, 98~
## $ chest_exhale_cm        <dbl> 98.6, 107.0, 98.0, 100.3, 104.5, NA, 81.0, 94.4~
## $ chest_expansion_cm     <dbl> 4.4, 2.5, 5.2, 4.2, 6.8, NA, 4.9, 4.1, 4.7, 6.2~
## $ lumbar_flexion_cm      <dbl> NA, 1.8, 4.1, 2.9, 6.3, NA, 2.8, 6.0, 3.3, 2.1,~
## $ age_years              <dbl> 34, 60, 26, 49, 42, 66, 45, 28, 44, 66, 49, 58,~
## $ sex                    <fct> Male, Female, Male, Female, Female, Male, Femal~
## $ race_ethnicity         <fct> Non-Hispanic White, Non-Hispanic Black, Mexican~
## $ pregnancy_status       <fct> NA, NA, NA, NA, Not pregnant, NA, NA, NA, NA, N~
## $ height_cm              <dbl> 164.7, 166.0, 173.0, 168.4, 164.3, 171.3, 166.7~
## $ weight_kg              <dbl> 87.4, 116.8, 97.6, 86.7, 107.7, 82.9, 75.7, 79.~
## $ bmi                    <dbl> 32.22, 42.39, 32.61, 30.57, 39.90, 28.25, 27.24~
## $ exam_weight            <dbl> 81528.772, 21000.339, 22633.582, 74112.487, 407~
## $ survey_psu             <dbl> 1, 2, 1, 2, 2, 1, 2, 1, 2, 2, 1, 2, 1, 1, 1, 2,~
## $ survey_stratum         <dbl> 83, 75, 88, 85, 75, 75, 78, 76, 79, 86, 82, 77,~
```

Step 6: Check the cleaned data

First, we confirm that the row count did not change, that IDs are still unique, and that the factor conversion did not lose any real answers. A value is only lost if it is `NA` after conversion but not before, so each number below should be 0.

```
nrow(clean_data) == nrow(raw_data)
```

```
## [1] TRUE
```

```
n_distinct(clean_data$id) == nrow(clean_data)
```

```
## [1] TRUE
```

```
c(
  back_pain         = sum(is.na(clean_data$back_pain) &
                          !is.na(recoded_data$ARQ010)),
  morning_stiffness = sum(is.na(clean_data$morning_stiffness) &
                          !is.na(recoded_data$ARQ040)),
  pain_with_rest    = sum(is.na(clean_data$pain_with_rest) &
                          !is.na(recoded_data$ARQ060)),
  exercise          = sum(is.na(clean_data$better_with_exercise) &
                          !is.na(recoded_data$ARQ080))
)
```

```
##         back_pain morning_stiffness    pain_with_rest          exercise
##                 0                 0                 0                 0
```

Next, we confirm that chest expansion equals the inhale measurement minus the exhale measurement. This should return `TRUE`.

```r
all.equal(
  clean_data$chest_expansion_cm,
  clean_data$chest_inhale_cm - clean_data$chest_exhale_cm,
  tolerance = 1e-6,
  check.attributes = FALSE
)
```

```
## [1] TRUE
```

Then we check the range of each measurement.

```r
summarize_measures(
  clean_data,
  c("occiput_wall_cm", "chest_expansion_cm", "lumbar_flexion_cm",
    "age_years", "height_cm", "weight_kg", "bmi")
)
```

```
## # A tibble: 7 x 6
##   measure                n missing   min median   max
##   <chr>              <int>   <int> <dbl>  <dbl> <dbl>
## 1 age_years           5001       0  20    44     69
## 2 bmi                 4962      39  15.0  28.2   84.9
## 3 chest_expansion_cm  4615     386   0     4.30  18.5
## 4 height_cm           4968      33 123.  167.   203.
## 5 lumbar_flexion_cm   4665     336   0     3.8   11
## 6 occiput_wall_cm     4938      63   0     0     16.5
## 7 weight_kg           4965      36  36.1  79.3  231.
```

Most participants have an occiput-to-wall distance of 0, which is the normal result, so this variable is very skewed. The highest chest expansion values stand out, so we list them.

```r
clean_data %>%
  arrange(desc(chest_expansion_cm)) %>%
  select(id, chest_inhale_cm, chest_exhale_cm, chest_expansion_cm) %>%
  head(10)
```

```
## # A tibble: 10 x 4
##       id chest_inhale_cm chest_exhale_cm chest_expansion_cm
##    <dbl>           <dbl>           <dbl>               <dbl>
##  1 54409            98.5            80                  18.5
##  2 59398           103.             87                  16.3
##  3 55384            88.9            73.2                15.7
##  4 56291            95              80                  15
##  5 59832            98.2            83.3                14.9
##  6 59300            94.6            80                  14.6
##  7 61231            99.2            85.4                13.8
##  8 60330            93              79.5                13.5
##  9 60715            94.7            81.6                13.1
## 10 60964            93.9            80.8                13.1
```

A few values are unusually high, but we cannot prove they are errors, so we keep them for now and will test whether they change our results during analysis.

Finally, we count the participants who were pregnant at the exam, since pregnancy can affect body measures and chest expansion.

```
count(clean_data, pregnancy_status)
```

```
## # A tibble: 4 x 2
##   pregnancy_status            n
##   <fct>                   <int>
## 1 Pregnant                   68
## 2 Not pregnant             1266
## 3 Could not be determined    43
## 4 <NA>                     3624
```

We keep these participants for now and record their status, so the team can decide whether to exclude them once the research question is final.

Step 7: Save the cleaned data

The cleaned data is saved so that the next file can load it without repeating these steps.

```
save(clean_data, file = "12_DataSaving.RData")
```